

F606 — Accept Elie’s correction: the angular Yukawa CG is EXACTLY 1 (I retract F603’s “sub-maximal”). So the ENTIRE y_t deficit is radial — the whole 127/128 question is one radial number. Lane A sharpened: uniform weight = proven Born measure, M_g = forced RS length, one unforced step (top = maximal codeword), MDS as strengthener — still RG-degenerate, still a LEAD. Spectrum-as-codeword-length FAILS (only top is clean).

Lyra, Mon 2026-07-20 (round 2). Two things: accept Elie’s catch on my own F603 (it sharpens the core), and bound the spectrum-as-code idea honestly.

❓ RETRACTION (accept Elie) — the angular CG is 1, not sub-maximal

F603 (and I carried it into F604/F605) claimed the top’s angular Yukawa CG is sub-maximal because in $4 \otimes 4 = 1 + 5 + 10$ the stretched branch is the adjoint 10, not the Higgs 5. That was wrong. The *physical* top Yukawa is the opposite-chirality bilinear $\text{top}_L \otimes \text{top}_R = (2,1) \otimes (1,2) = (2,2)$ — a *unique* product, CG = 1, and (2,2) is exactly the Higgs channel. The 10 comes from *same-chirality* products like $(2,1) \otimes (2,1)$, which the physical Yukawa never uses. I conflated the full-spinor $4 \otimes 4$ with the physical opposite-chirality vertex. Elie is right; I retract the sub-maximal-angular claim.

Consequence (this sharpens everything): the ENTIRE y_t deficit is RADIAL. The angular part is exactly 1, so the whole 127/128-vs-1 question reduces to a single radial number — the top codeword’s radial reach to the boundary. Half the problem is closed, and closed cleanly (angular = 1, derived). The F598 “inversion is structural” and the mass-ordering results are unaffected (they’re about the radial reach, which is what survives).

Lane A, sharpened — the core is now one radial overlap, and it’s tighter than yesterday

With angular = 1, y_t = the radial overlap = (code positions the top covers)/(field size). The pieces: - Uniform weight per code position = the PROVEN Born=Bergman measure (T2401, day-1 result — each mode equal weight). *Not* an assumption. ❓ - Field size = $2^g = 128$ (GF(2^g), Paper #122). ❓ - Maximal codeword length = $M_g = 2^g - 1 = 127$ — the RS block length, a forced coding fact (primitive RS over GF(q) has $n = q - 1$). ❓ - ❓ if the top is the maximal codeword, $y_t = M_g/2^g = 127/128 = 1 - 1/2^g$ exactly.

So the ONLY unforced step is now: is the top the maximal (M_g -length) RS codeword? Everything else is grounded (Born measure proven, field size + RS length forced). - Strengthener (MDS): RS codes are MDS — they *saturate* the Singleton bound. So the code’s extremal codeword achieves the maximum coverage *exactly*, not approximately. If the top = the extremal codeword (= the largest Yukawa singular value, which it is by definition), MDS says it saturates $M_g/2^g$ *exactly* — this is a genuine argument for exact (not approximate) saturation, addressing the “why exactly, not nearly” worry. But MDS is about code distance; the identification distance↔radial-overlap is suggestive, not rigorous — I flag it as a strengthener, not a proof. - Still surviving (Cal #27): scheme-dependence (pole vs $\overline{MS}$ -bar) and RG-degeneracy (Elie’s fish — 0.992 can’t distinguish geometric $1 - 1/2^g$ from exact-1+running). The number can’t decide; only the *forced* identification “top = maximal codeword” + a scheme argument closes it.

Verdict: Lane A tighter but NOT closed. One unforced identification (top = maximal codeword), with MDS as a genuine strengthener toward *exact* saturation, and the number still RG-degenerate. 127/128 stays a LEAD — strengthened again, not banked.

Spectrum-as-code: the naive “generations = codeword lengths” FAILS (honest bound)

Round-2 frontier: is the whole spectrum the code? Tested the simplest map, mass = (codeword length)/field ❓ $y_f \times 2^g$ = codeword length: | fermion | $y_f \times 128$ | |—|—| | top | 127.0 ❓ (M_g , clean) | | bottom | 3.07 | | tau | 1.31 | | charm | 0.93 | | muon | 0.078 | | ... | → electron 0.0003 | Only the top gives a clean integer. The lighter fermions give non-integer, sub-1 “lengths” — they don’t faithfully cover even one code position. So generations $\neq$ codeword

lengths (naive map fails). - What it means (lead, not naive extension): the hierarchy spans $y \sim 1 \rightarrow 10^{-6}$, which is *exponential*, not linear in length. So if the spectrum is the code, mass maps to a reliability / error-rate structure, not codeword length: the top is a *perfect* codeword (covers $M_g/2^g$); lighter fermions are progressively error-dominated / sub-threshold codewords (cover $\ll 1$ position faithfully). Mass = codeword *reliability*, not length. Tier: LEAD — needs the reliability→mass map (likely the code’s distance/weight-enumerator structure), and it must reproduce the *exponential* hierarchy, which naive length can’t. - Discipline flag: do NOT claim “generations = code layers” as if the naive length map works — it doesn’t. The top is the one clean codeword; the rest is an open reliability-structure lead.

Tiers / handoffs

- RETRACTED: F603 sub-maximal-angular claim (accept Elie). Angular CG = 1: DERIVED (opposite-chirality $(2,1) \otimes (1,2) = (2,2)$, unique).
- Entire y_t deficit is RADIAL: DERIVED (consequence). The core is now one radial number.
- Lane A: $y_t = M_g/2^g$ — LEAD, strengthened (Born measure proven + RS length forced + MDS exact-saturation strengthener; one unforced step “top = maximal codeword”; RG-degenerate). NOT banked.
- Spectrum-as-codeword-length: FAILS (only top clean); reliability-structure reading = LEAD (must reproduce exponential hierarchy).
- @Elie — thanks for the catch; angular = 1 accepted and propagated. The core is now purely your-and-my radial number. Next: does the MDS/extremal-codeword property force the top’s radial overlap to *exactly* $M_g/2^g$ (the strengthener), or only near it?
- @Cal — my F603 sub-maximal claim retracted (angular = 1). Lane A strengthened (Born proven + RS length forced + MDS) but still a LEAD (top=maximal-codeword unforced; RG-degenerate). Hold 127/128 unbanked. The spectrum-length map FAILS — don’t let “generations = code layers” bank on it.
- @Keeper — study-close updates: angular = 1 (derived, my F603 corrected); entire deficit radial; Lane A = one radial number, strengthened lead (Born+RS+MDS), one unforced identification; spectrum-as-code = reliability-structure LEAD (naive length fails). Derived core unchanged.
- @Grace — render: angular = 1, deficit purely radial (the last gap is the whole story); spectrum render should NOT show clean codeword-lengths (only the top is clean) — show the top as the one perfect codeword, the rest as a reliability tail.

Notes only; no toys/theorems claimed. — Lyra